

INTERPOLATION ERRORS THEOREM II

Let f be a function such that $f^{(n+1)}$ is continuous on $[a, b]$ and satisfies $|f^{(n+1)}(x)| \leq M$. Let p be the polynomial of degree $\leq n$ that interpolates f at $n + 1$ equally spaced nodes in $[a, b]$, including the endpoints. Then on $[a, b]$,

$$|f(x) - p(x)| \leq \frac{1}{4(n+1)} M \left(\frac{b-a}{n} \right)^{n+1} \quad (6)$$